Tensor

Free Space and Tensor Product

We are allowed to perform additions and multiplications on vectors. A natural question is whether we can perform a multiplication of two vectors. It is, however, unintuitive to imagine the result of such multiplication. Therefore, we would first define the **free vector space**, by just *stapling* two vectors together as the result of the multiplications. Then, we will see how to make sense of the result.

Free Space

Let V and W be two vector spaces with the field F. The **free vector space** arising from the **Cartesian product** of V and W is a vector space taking **every distinct pair of vectors** from V and W as the basis. It is denoted by $F(V \times W)$.
A typical element of the free vector space has the following form:

$$\boldsymbol{T} = \sum_{i=1}^{n} t_i (v_i, w_i),$$

given that $t_i \in F$, $v_i \in V$, and $w_i \in W$ for all i and that the pairs (v_i, w_i) are distinct.
Although the free vector space successfully defines the multiplication: the product of two vectors $v \in V$ and $w \in W$ is just the pair (v, w), it does not have any nice properties, such as

$$(av, w) = a(v, w),$$

for $a \in F$, $v \in V$, and $w \in W$. But in the free vector space, if $av \neq v$, then $a(v, w)$ is not the same as (av, w).

Bilinearity

Let's list out the desirable properties of the multiplication of two vectors:
1. Some sort of distributivity:
   1. $(v_1 + v_2, w) = (v_1, w) + (v_2, w)$
   2. $(v, w_1 + w_2) = (v, w_1) + (v, w_2)$
2. Property of scalar multiplication:
   1. $(av, w) = a(v, w)$
   2. $(v, aw) = a(v, w)$

Together, these properties are called **bilinearity**. In order to make these equalities hold, we take the **quotient space** of the free vector space by setting some elements to be 0.

Quotient Space and Subspace

To understand what is a **quotient space**, let's first start with a general vector space V and it's subset N. The quotient space of V by N, denoted by V/N, instead of containing the elements of V, contains the **cosets**. The coset of $v \in V$ is the set of all the elements of V that can be obtained by adding an element of N to v, denoted by $v + N$. In other words, the coset of v is the set

$$v + N = \{v + n \mid n \in N\}.$$

That is, V/N is the set of all $v + N$ for $v \in V$.
We also need to define the **equivalence relation** on V/N as follows: for $v_1 + N, v_2 + N \in V/N$, we say that $v_1 + N$ is equivalent to $v_2 + N$ if $v_1 - v_2 \in N$. In other words,

$$v_1 + N = v_2 + N \text{ if and only if } v_1 - v_2 \in N.$$

Now, one can show that if N is a **subspace** of V, then V/N is a vector space. To do so, we can define the addition and scalar multiplication of the quotient space as follows:

$$(v_1 + N) + (v_2 + N) = (v_1 + v_2) + N \text{ and } a(v + N) = (av) + N.$$

1. **Well-definedness of addition and scalar multiplication**:
   Since we **redefined** the **equivalence relation** on V/N, we need to show that the addition and scalar multiplication of the quotient space are **well-defined**, that is, if $v_1 + N, v_2 + N, v_1' + N, v_2' + N \in V/N$ and $a \in F$ such that $v_1 + N = v_1' + N$ and $v_2 + N = v_2' + N$, then
   $$(v_1 + v_2) + N = (v_1' + v_2') + N$$
   and
   $$(av) + N = (av') + N.$$
   To show this, consider
   $$(v_1 + v_2) - (v_1' + v_2') = (v_1 - v_1') + (v_2 - v_2').$$
   Since N is a subspace, we have $v_1 - v_1' \in N$ and $v_2 - v_2' \in N$, which implies that $(v_1 + v_2) - (v_1' + v_2') \in N$. This shows that
   $$(v_1 + v_2) + N = (v_1' + v_2') + N.$$
   Similarly, consider
   $$av - av' = a(v - v').$$
   Since N is a subspace, we have $v - v' \in N$, which implies that $a(v - v') \in N$. This shows that
   $$(av) + N = (av') + N.$$
   Hence, the addition and scalar multiplication of the quotient space are well-defined.
2. **Closure of addition and scalar multiplication**:
   We have to show that if $v_1 + N, v_2 + N \in V/N$ and $a \in F$, then
   $$(v_1 + v_2) + N \in V/N$$
   and
   $$(av) + N \in V/N.$$
   This is true because $v_1 + v_2 \in V$ and $av \in V$.
3. **Associativity of addition**:
   We have to show that if $v_1 + N, v_2 + N, v_3 + N \in V/N$, then
   $$v_1 + N + (v_2 + N + v_3 + N) = (v_1 + N + v_2 + N) + v_3 + N.$$
   We can show that both sides are equal to $(v_1 + v_2 + v_3) + N$.
   $$v_1 + N + (v_2 + N + v_3 + N) = v_1 + N + (v_2 + v_3) + N = (v_1 + v_2 + v_3) + N,$$
   and
   $$(V_1 + N + v_2 + N) + v_3 + N = (v_1 + v_2) + N + v_3 + N = (v_1 + v_2 + v_3) + N.$$
   This shows that the addition of the quotient space is associative.
4. **Commutativity of addition**:
   We have to show that if $v_1 + N, v_2 + N \in V/N$, then
   $$v_1 + N + v_2 + N = v_2 + N + v_1 + N.$$
   We can show that both sides are equal to $(v_1 + v_2) + N$.
   $$v_1 + N + v_2 + N = (v_1 + v_2) + N,$$
   and
   $$v_2 + N + v_1 + N = (v_2 + v_1) + N = (v_1 + v_2) + N.$$
   This shows that the addition of the quotient space is commutative.
5. **Existence of additive identity**:
   We have to show that there exists an element $0_{V/N} \in V/N$ such that for any $v + N \in V/N$, we have
   $$v + N + 0_{V/N} = v + N.$$
   This is true because $0 + N$ is a coset in V/N and
   $$v + N + 0 + N = (v + 0) + N = v + N.$$
6. **Existence of additive inverse**:
   We have to show that for any $v + N \in V/N$, there exists an element $-(v + N) \in V/N$ such that
   $$v + N + (-(v + N)) = 0_{V/N}.$$
   This is true because $-v + N$ is a coset in V/N and
   $$v + N + (-v) + N = (v - v) + N = 0 + N.$$
7. **Compatibility of scalar multiplication with field multiplication**:
   We have to show that if $a, b \in F$ and $v + N \in V/N$, then
   $$a(b(v + N)) = (ab)(v + N).$$
   This is true because
   $$a(b(v + N)) = a((bv) + N) = (abv) + N = (ab)(v + N).$$
8. **Distributivity of scalar multiplication with respect to vector addition**:
   We have to show that if $a \in F$ and $v_1 + N, v_2 + N \in V/N$, then
   $$a((v_1 + N) + (v_2 + N)) = a(v_1 + N) + a(v_2 + N).$$
   This is true because
   $$\begin{aligned} a((v_1 + N) + (v_2 + N)) &= a((v_1 + v_2) + N) \\ &= (a(v_1 + v_2)) + N = (av_1 + av_2) + N = a(v_1 + N) + a(v_2 + N). \end{aligned}$$
9. **Distributivity of scalar multiplication with respect to field addition**:
   We have to show that if $a, b \in F$ and $v + N \in V/N$, then
   $$(a + b)(v + N) = a(v + N) + b(v + N).$$
   This is true because
   $$(a + b)(v + N) = ((a + b)v) + N = (av + bv) + N = a(v + N) + b(v + N).$$

Tensor Product

Now, let's return to the free vector space $F(V \times W)$. We want to make the bilinearity properties hold. To do so, let's consider the set all the elements of the free vector space in the following forms:
1. $(v_1 + v_2, w) - (v_1, w) - (v_2, w)$
2. $(v, w_1 + w_2) - (v, w_1) - (v, w_2)$
3. $(av, w) - a(v, w)$
4. $(v, aw) - a(v, w)$

and **their linear combinations**, with $v, v_1, v_2 \in V$, $w, w_1, w_2 \in W$, and $a \in F$. This set is a subspace of the free vector space, denoted by N. Then, we can take the quotient space of the free vector space by N, denoted by $F(V \times W)/N$.
We showed that the quotient space of a vector space by it's subspace is a vector space. Therefore, $F(V \times W)/N$ is a vector space, denoted by $V \otimes W$, called the **tensor product** of V and W. Instead of writing (v, w), we write $v \otimes w$ to denote the product of $v \in V$ and $w \in W$.

Properties of Tensor Product

Decomposition of Tensors

Let V and W be two vector spaces with the field F. Let $\{v_i\}_{i \in I}$ and $\{w_j\}_{j \in J}$ be two bases of V and W, respectively. Then, the set $\{v_i \otimes w_j\}_{i \in I, j \in J}$ is a basis of the tensor product $V \otimes W$.
To show this, consider an element $t \in F(V \times W)/N$, from the definition of the quotient space,

$$t = t' + N$$

for some $t' \in F(V \times W)$. Thus, we can decompose t' as follows:

$$t' = \sum_{i=1}^{n} t_i' (v_i, w_i),$$

where $t_i' \in F$, $v_i \in V$, and $w_i \in W$ for all i and that the pairs (v_i, w_i) are distinct. Therefore,

$$t = \sum_{i=1}^{n} t_i' (v_i, w_i) + N$$

From the definition of vector addition in the quotient space, we have

$$t = \sum_{i=1}^{n} t_i' ((v_i, w_i) + N).$$

Then, since v_i and w_i can be decomposed as linear combinations of the basis vectors, we have

$$t = \sum_{i=1}^{n} t_i' \left(\left(\sum_{k \in I} a_{i,k} v_k, \sum_{l \in J} b_{i,l} w_l \right) + N \right),$$

From the bilinearity properties, we have

$$\begin{aligned} t &= \sum_{i=1}^{n} t_i' \left(\sum_{k \in I} \sum_{l \in J} a_{i,k} b_{i,l} (v_k, w_l) + N \right) \\ &= \sum_{i=1}^{n} \sum_{k \in I} \sum_{l \in J} t_i' a_{i,k} b_{i,l} ((v_k, w_l) + N) \end{aligned}$$

but since $v_k \otimes w_l$ is just $(v_k, w_l) + N$, we have

$$t = \sum_{i=1}^{n} \sum_{k \in I} \sum_{l \in J} t_i' a_{i,k} b_{i,l} v_k \otimes w_l.$$

This shows that any element of the tensor product can be decomposed as a linear combination of the basis vectors in $\{v_i \otimes w_j\}_{i \in I, j \in J}$.
In general, an element of the tensor product is written in the following form:

$$t = \sum_{i \in I} \sum_{j \in J} T^{ij} (v_i \otimes w_j).$$

Concrete Example of Tensor Product

Consider the vector space $\mathbb{R}^2$ with the standard basis $e_1 = (1, 0)$ and $e_2 = (0, 1)$. Then, the tensor product $\mathbb{R}^2 \otimes \mathbb{R}^2$ has the following basis:

$$e_1 \otimes e_1, e_1 \otimes e_2, e_2 \otimes e_1, e_2 \otimes e_2.$$

One can show that $\mathbb{R}^2 \otimes \mathbb{R}^2$ is isomorphic to $\mathbb{R}^4$ or a vector space of 2×2 real matrices.

Chained Tensor Products

Now, we have defined tensor as the product of two vectors, what about the product between a tensor and a vector or the product between two tensors?
Consider three vector spaces U, V, and W with the field F and the tensor product

$$(U \otimes V) \otimes W \text{ and } U \otimes (V \otimes W).$$

If $\{u_i\}_{i \in I}$, $\{v_j\}_{j \in J}$, and $\{w_k\}_{k \in K}$ are bases of U, V, and W, respectively, then the set $\{u_i \otimes v_j\}_{i \in I, j \in J}$ are the basis of $U \otimes V$. Therefore, a typical element of $(U \otimes V) \otimes W$ has the following form:

$$t = \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} T^{ijk} ((u_i \otimes v_j) \otimes w_k).$$

Similarly, we can define the tensor product $U \otimes (V \otimes W)$ and a typical element of it has the following form:

$$s = \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} S^{ijk} (u_i \otimes (v_j \otimes w_k)).$$

Consider the tensor product of u, v, and w, where $u \in U$, $v \in V$, and $w \in W$. We have

$$(u \otimes v) \otimes w = \left(\sum_{i \in I} a_i u_i \otimes \sum_{j \in J} b_j v_j \right) \otimes \sum_{k \in K} c_k w_k = \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} a_i b_j c_k ((u_i \otimes v_j) \otimes w_k)$$

and

$$u \otimes (v \otimes w) = \sum_{i \in I} a_i u_i \otimes \left(\sum_{j \in J} b_j v_j \otimes \sum_{k \in K} c_k w_k \right) = \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} a_i b_j c_k (u_i \otimes (v_j \otimes w_k)).$$

This shows that these tensor products are **canonically isomorphic**, that is, there is a **natural** isomorphism between $(U \otimes V) \otimes W$ and $U \otimes (V \otimes W)$ such that the tensor product of u, v, and w is the same in both spaces. Therefore,

$$(U \otimes V) \otimes W \cong U \otimes (V \otimes W).$$

Mulitlinearity

Since $(U \otimes V) \otimes W$ is isomorphic to $U \otimes (V \otimes W)$, the intuition is that we can somehow remove the parentheses and write the tensor product as $U \otimes V \otimes W$. However, notice that as $(U \otimes V) \otimes W$ and $U \otimes (V \otimes W)$ produce different mathematical objects, $U \otimes V \otimes W$ does not have a well-defined meaning. Therefore, we cannot just write $U \otimes V \otimes W$ without any clarification. We need a proper definition for multiple tensor products.

Consider the vector spaces $V_1, \cdots, V_n$ with the field F. We can define the **free vector space** arising from the Cartesian product of $V_1, \cdots, V_n$, denoted by $F(V_1 \times \cdots \times V_n)$, as the vector space taking every distinct n-tuple of vectors from $V_1, \cdots, V_n$ as the basis.
Again we are interested in the following properties of the multiplication of n vectors:
1. $(v_1, \cdots, x + y, \cdots, v_n) = (v_1, \cdots, x, \cdots, v_n) + (v_1, \cdots, y, \cdots, v_n)$
2. $(v_1, \cdots, kx, \cdots, v_n) = k(v_1, \cdots, x, \cdots, v_n)$

where $v_i \in V_i$ for all i; $x, y \in V_j$ for some j, and $k \in F$. The above properties are called **multilinearity**. With the same idea as the tensor product of two vector spaces, we can take the quotient space of the free vector space by the subspace N generated by all the elements in the above forms and their linear combinations. The resulting quotient space is called the **tensor product** of $V_1, \cdots, V_n$, denoted by $V_1 \otimes \cdots \otimes V_n$ or

$$\bigotimes_{i=1}^{n} V_i.$$

Since $F(V_1 \times \cdots \times V_n)/N$ is a quotient space of a vector space by its subspace, it is a vector space.

In the same way as the tensor product of two vector spaces, we can show that a typical element of the tensor product of n vector spaces has the following form:

$$t = \sum_{i_1 \in I_1} \sum_{i_2 \in I_2} \cdots \sum_{i_n \in I_n} T^{i_1 i_2 \cdots i_n} \left(v_{i_1}^1 \otimes v_{i_2}^2 \cdots \otimes v_{i_n}^n \right),$$

where $\{v_i^j\}_{i \in I_j}$ is a basis of V_j for all j. That is, the set

$$\left\{ v_{i_1}^1 \otimes v_{i_2}^2 \cdots \otimes v_{i_n}^n \right\}_{i_1 \in I_1, i_2 \in I_2, \cdots, i_n \in I_n}$$

is a basis of the tensor product of n vector spaces.
Now, consider $u \otimes v \otimes w$, where $u \in U$, $v \in V$, and $w \in W$. We have

$$u \otimes v \otimes w = \sum_{i \in I} a_i u_i \otimes \sum_{j \in J} b_j v_j \otimes \sum_{k \in K} c_k w_k = \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} a_i b_j c_k (u_i \otimes v_j \otimes w_k).$$

This shows that all three different tensor products, $(U \otimes V) \otimes W$, $U \otimes (V \otimes W)$, and $U \otimes V \otimes W$ are all **canonically isomorphic**, that is,

$$U \otimes V \otimes W \cong (U \otimes V) \otimes W \cong U \otimes (V \otimes W).$$

Moreover, as the result of the tensor product of u, v, and w generates the same coordinates for corresponding basis vectors in all three spaces, the **particular result** of the tensor product of u, v, and w is the same in all three spaces. Therefore, we can write

$$u \otimes v \otimes w \cong (u \otimes v) \otimes w \cong u \otimes (v \otimes w).$$

Thus, any operations performed between different tensor products of u, v, and w can be performed by first transforming the tensors through the isomorphic mapping into the same space and then performing the operations.
For instance, if we want to add elements between $(U \otimes V) \otimes W$ and $U \otimes (V \otimes W)$, we can first transform the elements in $(U \otimes V) \otimes W$ into $U \otimes (V \otimes W)$ through the isomorphic mapping and then perform the addition.

Simple Tensors

Consider the tensor product of n vector spaces $V_1, \cdots, V_n$ with the field F. A **simple tensor** is an element of the form $v_1 \otimes v_2 \cdots \otimes v_n$ where $v_i \in V_i$ for all i.

One can find **a sets of simple tensors** that spans the tensor product of n vector spaces. For instance, if $\{v_i^j\}_{i \in I_j}$ is a basis of V_j for all j, then the set

Universal Property of Tensor Product

The **universal property** of the tensor product states that for any vector space X and any **multilinear map** $f : V_1 \times \cdots \times V_n \mapsto X$, there exists a **unique linear map** $g : V_1 \otimes \cdots \otimes V_n \mapsto X$ such that

$$f = g \circ \varphi,$$

where $\varphi : V_1 \times \cdots \times V_n \mapsto V_1 \otimes \cdots \otimes V_n$ defined by $\varphi(v_1, \cdots, v_n) = v_1 \otimes \cdots \otimes v_n$ for all $v_i \in V_i$ for all i.
To prove this, consider the free vector space $F(V_1 \times \cdots \times V_n)$. Since f is a multilinear map, we can extend f to a linear map $\tilde{f} : F(V_1 \times \cdots \times V_n) \mapsto X$ by defining it as follows:

$$\tilde{f}\left(\sum_{i=1}^{k} t_k (v_k^1, v_k^2 \cdots, v_k^n) \right) = \sum_{i=1}^{k} t_k f(v_k^1, v_k^2 \cdots, v_k^n),$$

where $v_k^j \in V_j$ for all j and $t_k \in F$ for all k.
1. Let's first show that if $n \in N$, the subspace generated in the process of building the tensor product, then
   $$\tilde{f}(n) = 0.$$
   From the definition of N, we know that n is a linear combination of the following elements:
   1. $(v_1, \cdots, x + y, \cdots, v_n) - (v_1, \cdots, x, \cdots, v_n) - (v_1, \cdots, y, \cdots, v_n)$
   2. $(v_1, \cdots, kx, \cdots, v_n) - k(v_1, \cdots, x, \cdots, v_n)$

   where $v_i \in V_i$ for all i; $x, y \in V_j$ for some j, and $k \in F$. Let's represent those elements as e_i, then n can be written as
   $$n = \sum_{i=1}^{m} n_i e_i.$$
   Hence, we have
   $$\tilde{f}(n) = \tilde{f}\left(\sum_{i=1}^{m} n_i e_i \right) = \sum_{i=1}^{m} n_i \tilde{f}(e_i).$$
   However, for any particular e_i, we have
   $$\tilde{f}(e_i) = f((v_1, \cdots, x + y, \cdots, v_n)) - f((v_1, \cdots, x, \cdots, v_n)) - f((v_1, \cdots, y, \cdots, v_n)) = 0,$$
   or
   $$\tilde{f}(e_i) = f((v_1, \cdots, kx, \cdots, v_n)) - kf((v_1, \cdots, x, \cdots, v_n)) = 0$$
   because f is a multilinear map. Therefore, we have $\tilde{f}(n) = 0$.
2. Now we claim that the desire g is then given by
   $$g(t + N) = \tilde{f}(t),$$
   where $t \in F(V_1 \times \cdots \times V_n)$.
   Again, as we **altered** the **equivalence relation** on $F(V_1 \times \cdots \times V_n)/N = \bigotimes_{i=1}^{n} V_i$,we have to check whether g is well-defined. To do so, consider
   $$t_1 + N, t_2 + N \in \bigotimes_{i=1}^{n} V_i$$
   such that $t_1 + N = t_2 + N$. Then, we have $t_1 - t_2 \in N$.
   $$\begin{aligned} g(t_1 + N) &= g(t_2 + (t_1 - t_2) + N) \\ &= \tilde{f}(t_2 + (t_1 - t_2)) \\ &= \tilde{f}(t_2) + \tilde{f}(t_1 - t_2) \end{aligned}$$
   But $t_1 - t_2 \in N$, so $\tilde{f}(t_1 - t_2) = 0$. Therefore, we have
   $$g(t_1 + N) = \tilde{f}(t_2) = g(t_2 + N).$$
   This shows that g is well-defined.
3. Let's show that g satisfies the desired property, that is, for any $v_i \in V_i$ for all i, we have
   $$g \circ \varphi(v_1, \cdots, v_n) = f(v_1, \cdots, v_n).$$
   From the definition of φ and that g is well-defined, we have
   $$g \circ \varphi(v_1, \cdots, v_n) = g(v_1 \otimes \cdots \otimes v_n) = g((v_1, \cdots, v_n) + N) = \tilde{f}(v_1, \cdots, v_n) = f(v_1, \cdots, v_n).$$
   This shows that $g \circ \varphi = f$.
4. Let's show that g is a linear map. Consider $t_1 + N, t_2 + N \in \bigotimes_{i=1}^{n} V_i$ and $a \in F$. Then, we have
   $$\begin{aligned} g((t_1 + N) + (t_2 + N)) &= g((t_1 + t_2) + N) = \tilde{f}(t_1 + t_2) = \tilde{f}(t_1) + \tilde{f}(t_2) \\ &= g(t_1 + N) + g(t_2 + N), \end{aligned}$$
   and
   $$g(a(t + N)) = g((at) + N) = \tilde{f}(at) = a\tilde{f}(t) = ag(t + N).$$
   This shows that g is a linear map.
5. Let's show that g is the unique linear map such that $g \circ \varphi = f$. Suppose that there exists another linear map $g' : \bigotimes_{i=1}^{n} V_i \mapsto X$ such that $g' \circ \varphi = f$. Consider a **particular basis vector** of $\bigotimes_{i=1}^{n} V_i$, say
   $$v_1 \otimes v_2 \cdots \otimes v_n \in \left\{ v_{i_1}^1 \otimes v_{i_2}^2 \cdots \otimes v_{i_n}^n \right\}_{i_1 \in I_1, i_2 \in I_2, \cdots, i_n \in I_n},$$
   where $\{v_i^j\}_{i \in I_j}$ is a basis of V_j for all j. Then, we have
   $$g \circ \varphi(v_1, \cdots, v_n) = g(v_1 \otimes \cdots \otimes v_n) = f(v_1, \cdots, v_n) = g' \circ \varphi(v_1, \cdots, v_n) = g'(v_1 \otimes \cdots \otimes v_n).$$
   Since this holds for any basis vector of $\bigotimes_{i=1}^{n} V_i$ and both g and g' are linear maps, g must be the same as g'. Therefore, g is the unique linear map such that $g \circ \varphi = f$.

Dual Space

Dual Space of a Vector Space

The **dual space** of a vector space V with the field F, denoted by V^*, is the vector space of all linear maps from V to F. To show that V^* is a vector space, we can define the addition and scalar multiplication of V^* as follows: for $f, g \in V^*$ and $a \in F$,

$$(f + g)(v) = f(v) + g(v) \text{ and } (af)(v) = a(f(v)).$$

The axiom of vector spaces are automatically satisfied by V^* because the addition and scalar multiplication of V^* are defined in terms of the addition and scalar multiplication of F.

The **basis** of V^* is called the **dual basis** of the basis of V. To find a such basis, consider

$$v^*(v) = v^*\left(\sum_{i \in I} v_i e_i \right) = \sum_{i \in I} v_i v^*(e_i),$$

where $v \in V$, $v_i \in F$ for all i, and $\{e_i\}_{i \in I}$ is a basis of V and $v^* \in V^*$. Consider a linear combination of v^*, say

$$v^* = \sum_{j \in J} v_j^* f_j,$$

where $v_j^* \in F$ and $f_j \in V^*$ for all j. Then, we have

$$v^*(v) = \sum_{j \in J} v_j^* f_j \left(\sum_{i \in I} v_i e_i \right) = \sum_{j \in J} \sum_{i \in I} v_j^* v_i f_j(e_i).$$

Now the claim is that if we define f_j as follows:

$$f_j(e_i) = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases},$$

then the set $\{f_j\}_{j \in I}$ is a basis of V^*. To show this, we need to show that $\{f_j\}_{j \in I}$ is linearly independent and spans V^*.
1. $\{f_j\}_{j \in I}$ spans V^*:
   Since $v^*(e_j)$ are just scalars for all j, we define $v_j^* = v^*(e_j)$. Then, we have
   $$v^*(v) = \sum_{i \in I} v_i v^*(e_i) = \sum_{i \in I} \sum_{j \in I} v_i v_j^* f_j(e_i) = \sum_{j \in I} v_j^* f_j \left(\sum_{i \in I} v_i e_i \right) = \sum_{j \in I} v_j^* f_j(v).$$
   This shows that $\{f_j\}_{j \in I}$ spans V^*.
2. Independence of $\{f_j\}_{j \in I}$:
   First, let's show that the zero vector of V^* is the zero map, that is, the map that sends every vector in V to 0. For any $v^* \in V^*$ and $v \in V$, we have
   $$v^*(v) = (v^* + 0)(v) = v^*(v) + 0(v) = v^*(v) + 0$$
   This is enough to show that $0(v) = 0$ for all $v \in V$.
   Now, consider a linear combination of $\{f_j\}_{j \in I}$, say
   $$\sum_{j \in I} v_j^* f_j = 0,$$
   where $v_j^* \in F$ for all j. Then, the only possible way for the above linear combination to be the zero map is that $v_j^* = 0$ for all j. If some particular $v_k^* \neq 0$, then we have
   $$\sum_{j \in I} v_j^* f_j(e_k) = v_k^* f_k(e_k) = v_k^* \neq 0.$$
   This shows that $\{f_j\}_{j \in I}$ is linearly independent.

As $\{f_j\}_{j \in I}$ contains the same number of vectors as the basis of V, the dimension of V^* is the same as the dimension of V (the above proof is based on the assumption that V is finite-dimensional). Therefore, the dual space of V is isomorphic to V itself, that is, $V^* \cong V$.

Dual Space of a Dual Space

One can also define the dual space of the dual space of V, denoted by V^{**}. The elements of V^{**} are linear maps from V^* to F. One can show that V^{**} is **canonically isomorphic** to V itself, for this reason, we will practically treat V as the dual of V^*.

Contracting Tensors

Although we defined previously tensor products between two arbitrary vector spaces, we are particularly interested in the tensor product between a vector space and its dual space, that is, $V \otimes V^*$. This is useful because it allows us to define the **contraction** of a tensor.

Let's start with a concrete example, consider $\mathbb{R}^2 \otimes (\mathbb{R}^2)^*$. This particular tensor product is called **rank(1,1) tensor**, in general, a tensor product

$$T_n^m(V) = \underbrace{V \otimes \cdots \otimes V}_{m} \otimes \underbrace{V^* \otimes \cdots \otimes V^*}_{n}$$

is called **rank(m,n) tensor**.
Let e_1, e_2 be the standard basis of $\mathbb{R}^2$ and e^1, e^2 be the dual basis of e_1, e_2, that is

$$e^i(e_j) = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{if } i \neq j \end{cases},$$

condensed notation for the above is $e^i(e_j) = \delta_j^i$, called the **Kronecker delta**.
Then, the set

$$\{e_i \otimes e^j\}_{i,j=1,2}$$

are the basis of $\mathbb{R}^2 \otimes (\mathbb{R}^2)^*$. Thus, a typical element of $\mathbb{R}^2 \otimes (\mathbb{R}^2)^*$ has the following form:

$$T = \sum_{i=1}^{2} \sum_{j=1}^{2} T_j^i (e_i \otimes e^j),$$

or, in **Einstein summation convention**, we can write

$$T = T_j^i (e_i \otimes e^j).$$

Here, the **Einstein summation convention** is a notation that we will sum over any index appeared both as upper and lower indices. For instance, $v_i e^i = \sum_{i=1}^{n} v_i e^i$.
We know that $\mathbb{R}^2 \otimes (\mathbb{R}^2)^*$ is isomorphic to the vector space of 2×2 real matrices. Hence, we can represent T as a 2×2 matrix as follows:

$$T = T_j^i (e_i \otimes e^j) \cong \begin{bmatrix} T_1^1 & T_2^1 \\ T_1^2 & T_2^2 \end{bmatrix} = \boldsymbol{T}.$$

The **contraction** of T evaluates the tensor product by the duality between $\mathbb{R}^2$ and $(\mathbb{R}^2)^*$. In particular, the contraction of $T = T_j^i (e_i \otimes e^j)$ is defined as follows:

$$C(T) = C\left(T_j^i (e_i \otimes e^j) \right) = T_j^i C(e_i \otimes e^j) = T_j^i e^j(e_i).$$

Since $e^j(e_i) = \delta_i^j$, the only terms that survive in the above summation are those with $i = j$. Therefore, we have

$$C(T) = T_i^i = \sum_{i=1}^{2} T_i^i = \operatorname{tr}(\boldsymbol{T}).$$

The contraction of T is just the **trace of the matrix** representation of T.

Contraction of Tensors

The **contraction operator** C_l^k acts on a rank(m,n) **simple tensor**

$$v = \left(v^1 \otimes \cdots \otimes v^m \otimes v_1 \otimes \cdots \otimes v_j \right)$$

by evaluating the k-th upper index and the l-th lower index. In particular, we have

$$C_l^k(v) = v^k(v_l) \left(v^1 \otimes \cdots \otimes \widehat{v^k} \otimes \cdots \otimes v^m \otimes v_1 \otimes \cdots \otimes \widehat{v_l} \otimes \cdots \otimes v_j \right),$$

where $\widehat{v^k}$ and $\widehat{v_l}$ means that v^k and v_l are removed from the tensor product.
Let's define **the contraction of a general rank(m,n) tensor**. Consider a general rank(m,n) tensor, as simple tensors span the tensor product, we can decompose the general tensor as a linear combination of simple tensors S_i. In particular, we have

$$C_l^k(T) = C_l^k\left(\sum_i T_i S_i \right) = \sum_i T_i C_l^k(S_i).$$

We can also decompose the general tensor as a linear combination of basis vectors, say

$$T = T_{j_1 \cdots j_n}^{i_1 \cdots i_m} \left(e_{i_1} \otimes \cdots \otimes e_{i_m} \otimes e^{j_1} \otimes \cdots \otimes e^{j_n} \right),$$

where $\{e_i\}_{i \in I}$ is a basis of V and $\{e^j\}_{j \in I}$ is the dual basis of $\{e_i\}_{i \in I}$. Then, we have

$$\begin{aligned} C_l^k(T) &= T_{j_1 \cdots j_n}^{i_1 \cdots i_m} C_l^k \left(e_{i_1} \otimes \cdots \otimes e_{i_m} \otimes e^{j_1} \otimes \cdots \otimes e^{j_n} \right) \\ &= T_{j_1 \cdots j_n}^{i_1 \cdots i_m} e^{j_k}(e_{i_l}) \left(e_{i_1} \otimes \cdots \otimes \widehat{e_{i_l}} \otimes \cdots \otimes e_{i_m} \otimes e^{j_1} \otimes \cdots \otimes \widehat{e^{j_k}} \otimes \cdots \otimes e^{j_n} \right). \end{aligned}$$

The only terms that survive in the above summation are those with $i_l = j_k$. Therefore, we have

$$T_{j_1 \cdots j_n}^{i_1 \cdots i_m} e^{j_k}(e_{i_l}) = T_{j_1 \cdots j_n}^{i_1 \cdots i_m} \delta_{i_l}^{j_k} = \sum_{i_l = j_k} T_{j_1 \cdots j_n}^{i_1 \cdots i_m}.$$

Thus, the contraction of T is just the summation of the coordinates of T with $i_l = j_k$.

$$C_l^k(T) = \sum_{i_l = j_k} T_{j_1 \cdots j_n}^{i_1 \cdots i_m} \left(e_{i_1} \otimes \cdots \otimes \widehat{e_{i_l}} \otimes \cdots \otimes e_{i_m} \otimes e^{j_1} \otimes \cdots \otimes \widehat{e^{j_k}} \otimes \cdots \otimes e^{j_n} \right).$$

Matrix Multiplication as Contraction of Tensors

This seems a bit abstract, let's do some concrete examples to see how the contraction of a tensor works. Consider two tensors from $\mathbb{R}^2 \otimes (\mathbb{R}^2)^*$, say

$$T = T_j^i (e_i \otimes e^j) \cong \begin{bmatrix} T_1^1 & T_2^1 \\ T_1^2 & T_2^2 \end{bmatrix} = \boldsymbol{T} \text{ and } S = S_j^i (e_i \otimes e^j) \cong \begin{bmatrix} S_1^1 & S_2^1 \\ S_1^2 & S_2^2 \end{bmatrix} = \boldsymbol{S}.$$

If we perform the tensor product between T and S, we have

$$T \otimes S = T_j^i S_l^k (e_i \otimes e^j \otimes e_k \otimes e^l).$$

Let's apply the contraction operation at the middle two indices, that is, we evaluate the last upper index and the first lower index. Then, we have

$$\begin{aligned} C_1^2(T \otimes S) &= T_j^i S_l^k C_1^2 (e_i \otimes e^j \otimes e_k \otimes e^l) \\ &= T_j^i S_l^k e^j(e_k) (e_i \otimes \widehat{e^j} \otimes \widehat{e_k} \otimes e^l) \\ &= T_j^i S_l^k \delta_k^j (e_i \otimes e^l) \\ &= \sum_{j=k} T_j^i S_l^k (e_i \otimes e^l) \\ &= (T_1^1 S_1^1 + T_2^1 S_1^2)(e_1 \otimes e^1) + (T_1^1 S_2^1 + T_2^1 S_2^2)(e_1 \otimes e^2) \\ &\quad + (T_1^2 S_1^1 + T_2^2 S_1^2)(e_2 \otimes e^1) + (T_1^2 S_2^1 + T_2^2 S_2^2)(e_2 \otimes e^2) \\ &\cong \begin{bmatrix} T_1^1 S_1^1 + T_2^1 S_1^2 & T_1^1 S_2^1 + T_2^1 S_2^2 \\ T_1^2 S_1^1 + T_2^2 S_1^2 & T_1^2 S_2^1 + T_2^2 S_2^2 \end{bmatrix} = \boldsymbol{TS}. \end{aligned}$$

Notice how the contraction of $T \otimes S$ is just the matrix multiplication of the matrix representation of T and the matrix representation of S.

Basis Change of Tensors

As in vectors, we can also perform basis change for tensors. Let's first discuss what happens when we perform basis change for the vector space and its dual space.
Let V be a vector space with the field F and $\{e_i\}_{i \in I}$ be a basis of V. Then, the dual basis of $\{e_i\}_{i \in I}$ is $\{e^j\}_{j \in I}$, where

$$e^j(e_i) = \delta_i^j.$$

Now, let's consider another basis of V, say $\{f_i\}_{i \in I}$. A vector $v \in V$ can be represented in both bases as follows:

$$v = v^i e_i = v'^i f_i.$$

Let's say that for a particular i, we have $e_i = E_i^j f_j$. Then, we have

$$v^i e_i = v^i E_i^j f_j = v'^i f_i.$$

Hence,

$$v'^i = v^j E_j^i.$$